

SRIRAM M

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Generative AI, LLM applications, and production-grade machine learning systems. Experienced in building scalable AI solutions including RAG pipelines, multi-agent systems, LLM fine-tuning, and computer vision models using LangChain, LangGraph, FAISS/Qdrant, PyTorch, and YOLOv8. Skilled across the end-to-end ML lifecycle — data engineering, model development, evaluation, and deployment — using FastAPI, Docker, Kubernetes, AWS, and MLOps/CI-CD practices. Proficient with MCP-based tool calling, structured outputs, and agentic orchestration for reliable, high-performance AI systems.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Optimization (GridSearch, Optuna)

Deep Learning: PyTorch, TensorFlow, Neural Networks, Computer Vision, NLP

Generative AI & Agents: LLM Applications, RAG, Transformer Architecture, Attention Mechanisms, LangChain, LangGraph, LlamaIndex, Prompt Engineering, AI Agents, LoRA/QLoRA, MCP (Model Context Protocol), Function Calling, Structured Outputs, Tool Calling, n8n

LLM Serving: vLLM, Ollama, TensorRT-LLM, Hugging Face Inference

Vector Databases: FAISS, Qdrant, Pinecone, Embeddings

Databases: PostgreSQL, MongoDB

Backend & APIs: FastAPI, SQLAlchemy, JWT, OAuth2, WebSockets, Pydantic, REST API Design

MLOps & Deployment: Docker, Kubernetes, MLflow, Redis, Celery, CI/CD, Git, Model Monitoring, Data Drift Detection

Cloud (AWS): EC2, Bedrock, SageMaker, ECS, Lambda, S3

Evaluation & Testing: RAGAS, DeepEval, TruLens, Promptfoo, Pytest, Unit & Integration Testing

Architecture & Practices: Clean Architecture, SOLID Principles, Design Patterns, Microservices, System Design

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern(On-Site)

Dec 2025 – Jun 2026

- Built and deployed an NLP-based customer support chatbot using FastAPI and LangChain, reducing average query resolution time by ~40% across 500+ daily user interactions.
- Developed a predictive analytics model for business forecasting using XGBoost; achieved AUC-ROC of 0.91 after iterative feature engineering on a 150K-row dataset.
- Designed and implemented REST API endpoints for real-time ML model inference, supporting concurrent requests with sub-200ms response latency using async FastAPI.
- Automated a data preprocessing and model retraining pipeline using Celery and Redis, reducing manual intervention by ~60% and cutting pipeline turnaround time from 4 hours to 45 minutes.
- Containerized AI applications using Docker and deployed to AWS EC2, enabling consistent staging-to-production delivery across 3 internal business use cases.
- Collaborated in code reviews, wrote unit tests (Pytest), and maintained CI/CD pipelines for ML model versioning and deployment using Git and MLflow.

PROJECT EXPERIENCE

Enterprise RAG + Agentic Knowledge Assistant

- Built a production-grade RAG system using LangChain, LangGraph, and FAISS/Qdrant with agent-based query routing between retrieval and direct LLM responses.
- Implemented hybrid search (Hugging Face embeddings + BM25 + cross-encoder reranking) with RRF fusion and HyDE query transformation, improving retrieval precision by 31%.
- Evaluated using RAGAS metrics and deployed as an async FastAPI + Docker + AWS service with Redis caching and streaming responses.

Fine-Tuned LLM Microservice (LoRA/QLoRA)

- Fine-tuned Llama-3.1-8B for text-to-SQL generation using PEFT LoRA/QLoRA, reducing trainable parameters by 94% while achieving 88% execution accuracy.
- Applied 4-bit quantization (bitsandbytes) to optimize model size and inference performance; tracked experiments with MLflow.
- Deployed optimized inference APIs using FastAPI, Docker, and CI/CD, with vLLM-based serving for higher throughput.

Multi-Agent Autonomous AI Workflow Platform

- Designed a multi-agent AI automation platform using LangGraph (planner/critic/executor architecture) with specialized agents for classification, RAG retrieval, response generation, and escalation.
- Implemented memory, MCP-based tool calling, API integrations, structured outputs, and human-in-the-loop approval.
- Deployed backend using FastAPI, Docker, and Redis with a Streamlit monitoring dashboard.

Customer Churn Prediction ML System (XGBoost)

- Built an end-to-end ML pipeline to predict customer churn from structured business data, including preprocessing, feature engineering, and class-imbalance handling.
- Compared Logistic Regression, Random Forest, and XGBoost; achieved best performance with XGBoost after hyperparameter tuning (GridSearch/Optuna).
- Deployed the model through a monitored FastAPI service with Pytest-based test coverage.

Real-Time Computer Vision + MLOps Pipeline

- Developed an industrial defect detection system using YOLOv8 and PyTorch, achieving 93% mAP on a custom 10K-image dataset.
- Built real-time inference APIs with FastAPI, Redis, and Celery, deployed with Kubernetes on AWS.
- Implemented full MLOps workflow including experiment tracking, model versioning, A/B testing, and drift monitoring.

CERTIFICATIONS

- AI Coder: Vibe Coder to Agentic Engineer — Udemy, 2025
- Deep Learning & Reinforcement Learning — Coursera, 2024
- Generative AI Models & Platforms — IBM / Coursera, 2024
- Getting Started with DevOps on AWS — AWS Training, 2024
- Python Programming — GeeksforGeeks, 2023
- Prompt Engineering Basics — Coursera, 2024

EDUCATION

Master of Computer Applications (MCA) <i>GPA: 8.4 / 10</i>	2024 – 2026
Bachelor of Computer Applications (BCA) <i>GPA: 7.0 / 10</i>	2021 – 2024

ADDITIONAL INFORMATION

Languages: English (Fluent), Tamil (Native), Kannada (Conversational)
Interests: Traveling, Fitness, Coding, MotoGP, Formula 1